

Co-Route Fiber Recognition and Status Diagnosis Based on Integrated Sensing and Communication in 6G Transport Networks

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Abstract—The 6G transport network facilitates the Internet of Everything (IoE), carrying numerous services and emphasizing the paramount importance of its reliability. However, within the transport network, the issue of co-route fibers arises. The co-route fibers, encompassing both co-cable and co-trench fibers, presents a significant latent hazard for service disruptions, posing a substantial threat to the seamless connectivity envisioned for the 6G era of pervasive IoE. The segregation of communication and sensing in the transmission network results in mutual interference between communication and sensing signals, rendering it difficult to promptly address sudden fiber interruptions. This article proposes an integrated sensing and communication (ISAC) architecture within transport networks, aiming at the online discernment of co-cable fibers, characterization of fiber optic trenches, and real-time classification of fiber vibration events. In the domain of co-cable fiber identification, our approach has successfully reduced the nuisance alarm rate to an impressive 5.3%, while simultaneously elevating the recognition accuracy to an outstanding 99.7%. As for co-trench fiber identification, our proposed methodology not only facilitates the discernment of co-trench fibers but also achieves an impressive accuracy of 97.7% in classifying fiber trenches. Moreover, in the realm of fiber state prediction, our solution has achieved a remarkable recognition accuracy of 98% across six distinct vibration events. These results underscore the robust performance of the proposed ISAC architecture, which will effectively safeguard the survivability of 6G IoE.

Index Terms—Co-route fiber recognition, distributed optical fiber sensing, event classification, feature extraction algorithm (FEA), machine learning (ML) models.

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I. INTRODUCTION

IN THE era of 6G, network offers faster data transmission speeds and lower latency, enhancing virtual and augmented reality experiences and enabling scenarios, such as remote healthcare and smart homes [1], [2]. In the realm of 6G, the transport network serves as its foundational infrastructure, playing a pivotal role in the overall system [3]. Through the transport network, individuals, sensors [4], [5], remote devices, and environmental information are seamlessly integrated into the network, culminating in the Internet of Everything (IoE) [6], [7]. IoE facilitates comprehensive information exchange among various devices and objects, thereby delivering significant convenience to human life and work [8], [9].

In the IoE, scenarios like smart transportation, autonomous driving, and smart cities require extremely low latency and high reliability [10], [11], [12]. Any interruption in network connectivity could significantly impede the development of IoE, resulting in substantial economic and societal losses. According to a rough estimate cited by Reuters, global profit-generating websites suffer a loss of \$29 million in revenue for every hour of interruption [13]. In the transport network, the passive nature of optical fibers may lead to the deployment of primary and backup routes for certain services within the same trench or cable, resulting in the phenomenon of co-route fibers [14]. If the co-route fiber experiences fiber abrasion fault, it will directly result in service interruption. Hence, monitoring the status of optical fibers and identifying fiber route redundancy is of paramount importance.

The phenomenon of co-route fibers encompasses both co-cable and co-trench fibers. To discern co-cable fibers, neural networks were leveraged in [15] to acquire weights for computing the similarity of optical fiber events. Reference [16], on the other hand, identified vibration features from data streams to detect fibers within the same cable. Furthermore, in [14], an artificial intelligence (AI) method based on deep siamese neural networks was proposed for detecting co-route optical fibers. The abrupt nature of network failures necessitates real-time responses from network administrators, thus demanding fault detection models with robust generalization capabilities. While commendable contributions have been made by these

endeavors, they still fall short in terms of online recognition and robust generalization capabilities.

Optical time-domain reflectometers (OTDRs) are commonly used for fiber detection [14]. However, traditional OTDR instruments interfere with communication signals, rendering them incapable of achieving online monitoring. The coexistence of communication and sensing signals in transport networks promises significant convenience for real-time perception and communication coordination. To meet the low-latency, high-reliability demands of the IoE, it is imperative to leverage integrated sensing and communication (ISAC) in transport networks to identify co-route fiber. In our previous work [13], we facilitated online recognition of co-route fiber through real-time interaction between intelligent sensing units (ISUs) and the intelligent recognition unit (IRU). But this scheme lacks the capability to provide advance warning for sudden behaviors that may damage optical cables.

This article presents an ISAC architecture in transport network, which capable of online co-cable fiber identification, optical cable trench characterization, and fiber vibration event recognition. This architecture enables real-time warning of optical cable status and identification of co-route fiber. The architecture begins by constructing a feature extraction algorithm (FEA) to extract the signal's time-frequency domain characteristics. In the time domain, statistical properties of the time series are extracted, while in the frequency domain, features are extracted using fast Fourier transform (FFT) and wavelet packet decomposition (WPD). This FEA used in this work is denoted as FEA-TFW (FEA with Time, FFT, and WPD). The ISU extracts both dynamic and static sensory data from the fiber and utilizes the custom-built FEA-TFW to extract feature vectors (FVs). These FVs are then transmitted in real time to the IRU. Subsequently, the IRU employs feature dimension reduction (FDR) algorithms to reduce the dimensionality of the FVs, which are then fed into pretrained machine learning (ML) models to accomplish the recognition tasks for co-route fibers and fiber vibration event types.

We collected a significant amount of fiber sensing data from the operator's live network environments and validated the performance of our proposed architecture using open available data sets of fiber vibration events. Experimental results demonstrate a co-cable fiber identification accuracy of 99.7%, optical fiber trench identification accuracy of 97.7%, and fiber vibration event identification accuracy of 98%. Its key points lie in the following aspects.

- 1) In the realm of co-cable fiber identification, our proposed framework introduces a novel approach by considering the fused OTDR curves of fiber pairs as the fundamental unit for model processing. This method of feature construction imbues the FV with inherent interfiber pair correlations from the outset, thereby elevating the co-cable recognition accuracy to 99.7%.
- 2) In the aspect of co-trench fiber identification, the architecture proposed in this article characterizes different categories of fiber trenches. This approach offers the advantage of not only identifying whether the fibers to be identified belong to the same trench but also determining the specific trench they originate from.

- 3) In the field of optical fiber detection and operations, a novel architecture is introduced for the first time, capable of simultaneously identifying co-route fiber and fiber vibration events. By leveraging integrated sensing in transport networks to identify risks associated with co-route fiber and providing real-time warnings on fiber status, this architecture enhances the survivability of 6G IoE services in optical networks.

The rest of this article is organized as follows. Section II summarizes the research status of optical fiber sensing and ISAC in transport network. Section III introduces the system architecture and mathematical model of FEA and FDR. Section IV presents the results of simulation experiments. Finally, conclusions are given in Section V.

II. RELATED WORK

The advancement of 6G has drawn researchers' attention to enhancing the capacity [17], [18], [19] and efficiency [20], [21], [22] of communication systems. The improvement in capacity and efficiency necessitates a highly reliable transport network to ensure communication quality [23], [24], [25]. In efforts to enhance the survivability of transport networks, researchers have dedicated considerable efforts to path planning [26], [27], [28], spectrum allocation [29], [30], [31], and network device fault detection [32], [33]. However, these endeavors have not fundamentally addressed the issue of co-route fiber. It is crucial to identify co-route fiber based on the characteristics of optical fibers themselves. Extracting the characteristics of optical fibers necessitates the use of the "probe radar" within the fiber—OTDR, which utilizes intensity of backscattered signals to collect information about events such as loss and bending in optical fibers, making it widely used for fault localization in engineering practice [34].

In recent years, researchers have shown a keen interest in applying AI algorithms, such as ML and deep learning (DL) models to the field of OTDR sensing. In [35], researchers combined autoencoder-based anomaly detection with attention-based bidirectional gated recurrent unit algorithms, significantly reducing the time and cost of network operations and maintenance. In [36], DL models were used to automatically detect optical events in OTDR traces. Thanks to a specialized preprocessing pipeline, this model can even identify events not present in the training data. Additionally, in [37], it was demonstrated for the first time that OTDR distance can accurately determine the geographic location of deployed fiber cables. With an accuracy of 4 m for buried cables, this approach will greatly enhance the efficiency of operational teams in locating and repairing fiber issues in the field.

In addition to intensity, the phase variation of Brillouin scattered signals in optical fibers can be utilized as a sensing parameter through coherent detection techniques, enabling the measurement of extremely subtle vibrations at any location along tens of kilometers of fiber [38].

This type of OTDR, which detects the phase of the backscattered echo, is referred to as φ -OTDR [39]. Due to its high sensitivity and low cost, it has been widely

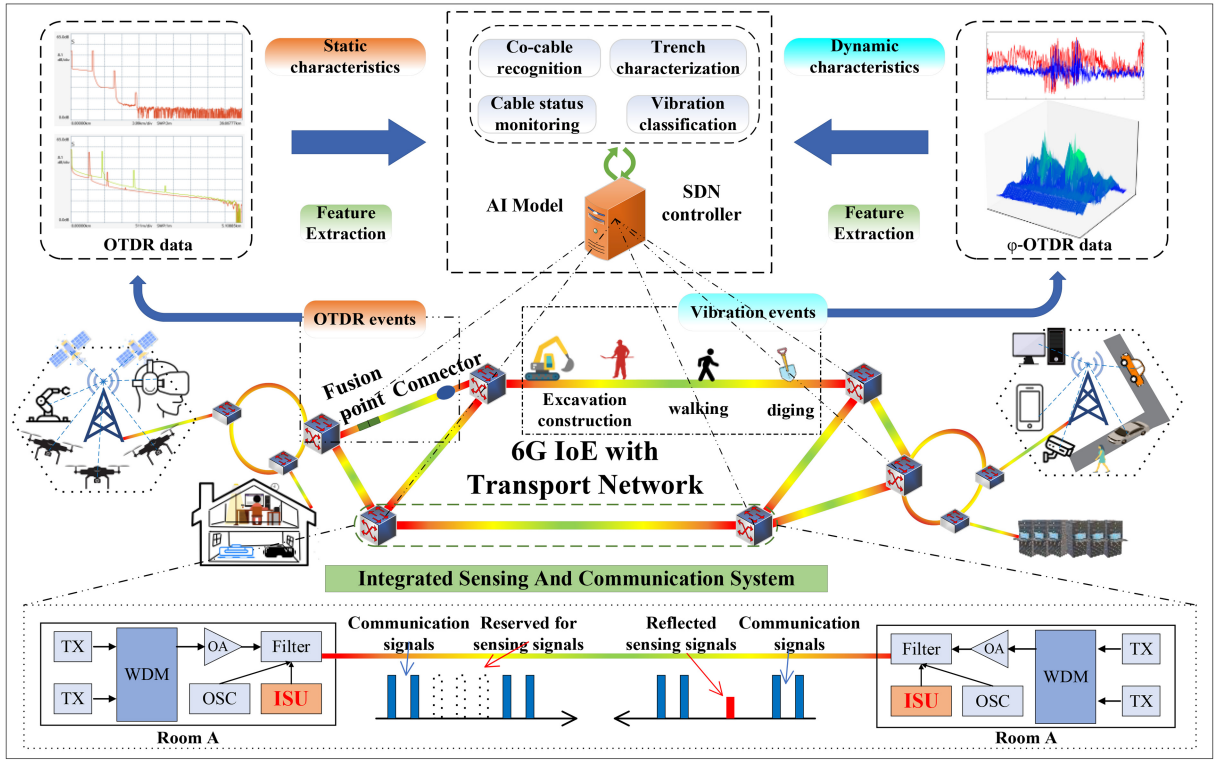


Fig. 1. Architecture diagram of fiber identification and condition warning in transport networks based on ISAC in 6G IoE.

employed in various fields, such as perimeter security surveillance [40], geological exploration [41], and pipeline monitoring [42]. However, the high nuisance alarm rate (NAR) and missed detection issues under strong interference remain significant challenges that need to be addressed before widespread application of φ -OTDR technology [43]. To enhance the performance of φ -OTDR in vibration pattern recognition, researchers have extensively validated ML and DL models [44].

Traditional ML models offer faster training and prediction speeds [45]. To accomplish real-time tasks, it is essential not only for the algorithm model to meet the requirements but also for the system hardware to be appropriately adapted. Through improvements, OTDR can now detect optical fiber links without affecting existing network operations [13]. The integration of this technology can be seen as the embryo of ISAC in transport network. Nowadays, a significant deployment of communication optical fibers has been observed globally, and the reuse of existing communication optical fibers for integrated sensing in transport networks has garnered considerable attention [46], [47].

At the signal level, researchers have unveiled a novel avenue for the ISAC in transport network by harnessing linear frequency modulated light as the optical carrier for pulse amplitude modulation 4-Level signal transmission and employing Φ -OTDR as the sensing probes [48]. Reports on the ISAC in transport network using space-division multiplexing (SDM) appeared in 2023 [49], where researchers deployed a cable containing uncoupled seven-core fibers in underground tunnels. They successfully demonstrated high-capacity SDM

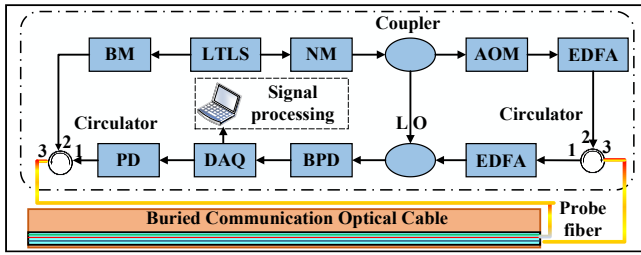
transmission and conducted real-time monitoring of urban traffic. In 2020, the first field trial of distributed fiber optic sensing (DFOS) and high-speed communication was reported on an operational telecom network [50]. This solution demonstrated the feasibility of ISAC in transport network based on wavelength division multiplexing systems.

Up to now, integrated sensing solutions in transport networks have not been applied to the field of co-route fiber detection. In this article, we leverage an ISAC architecture in transport networks to achieve the identification of co-route fiber and real-time warning of fiber status, building upon previous work. This holds significant implications for the realization of 6G IoE.

III. SYSTEM ARCHITECTURE AND MATHEMATICAL MODEL

A. System Architecture

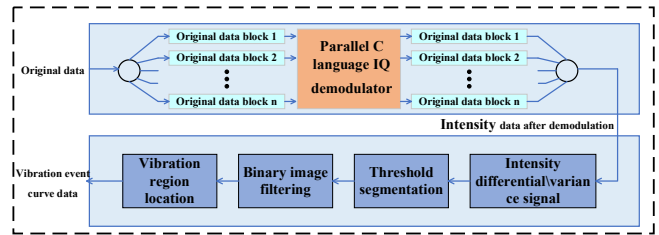
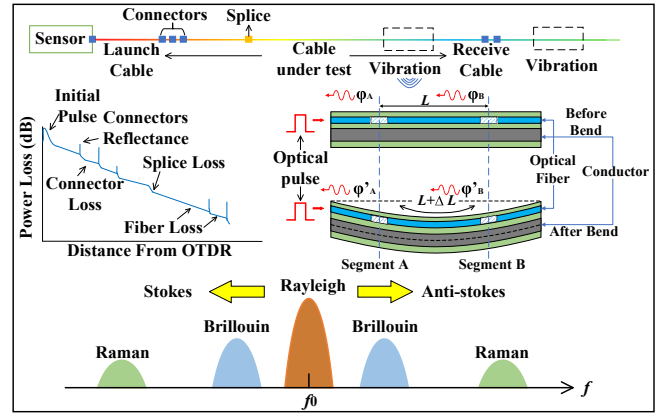
In Fig. 1, we illustrate the ISAC architecture proposed for transport networks. In the context of 6G IoE, optical networks serve as a crucial foundation for the IoE, where optical fibers within the network act not only as waveguides for communication signals but also as sensors for perception parameters. In this article, we implement DFOS using ISUs embedded in transport network devices to extract sensing information from optical fibers without interrupting existing communication services. The task of identifying co-route fiber is performed by the IRU, which is deployed within the software-defined networking (SDN) controller. The IRU contains a pretrained recognition model and completes the



As shown in Fig. 3, the backscattering of optical fibers includes Rayleigh, Brillouin, and Raman scattering [50]. OTDR can extract connectors and splicing losses through Rayleigh backscattering and Fresnel reflection, with its physical expression represented by

When the LTLS operates in NM mode, the ISU analyzes the vibration information of the optical fiber through phase difference, which is given by

By measuring the phase change of the returned optical signal, it is possible to analyze vibration characteristics within the optical fiber, such as the location, frequency, and amplitude of the vibration.



In traditional ML algorithms, performance relies on effective FEA. Before performing feature extraction, it is necessary to normalize the raw data. In this article, we employ the Min-Max scaling method for data normalization, which is represented by the following formula:

In the time domain, 15 features are extracted from the normalized signal, as shown in [45, Table IV]. These features include maximum value, minimum value, peak-to-peak value,

TABLE I
FEATURES IN TIME DOMAIN

Maximum (<i>Max</i>)	Minimum (<i>Min</i>)	Peak-to-peak value (<i>PK-PK</i>)
Mean (<i>A</i>)	Variance (<i>var</i>)	Standard deviation (<i>SD</i>)
Time domain entropy (<i>TDE</i>)	Root mean square (<i>RMS</i>)	Average rectified value (<i>Arv</i>)
Shape factor (<i>Sf</i>)	Pulse factor (<i>Pf</i>)	Crest factor (<i>Cf</i>)
Clearance factor (<i>CL</i>)	Kurtosis factor (<i>Kf</i>)	Skewness factor (<i>Skew</i>)

TABLE II
FEATURES ASSOCIATED WITH FFT

Spectral energy (<i>Se</i>)	Spectral domain entropy (<i>SDE</i>)
Dominant frequency (<i>DF</i>)	Mean Frequency (<i>MF</i>)
Spectral shape factor (<i>SSf</i>)	Spectral Slope (<i>SS</i>)
Spectral peak factor (<i>SPf</i>)	

TABLE III
FEATURES ASSOCIATED WITH WPD

LLP energy (<i>LLPE</i>)	LHP energy (<i>LHPE</i>)	HLP energy (<i>HLPE</i>)
HHP energy (<i>HHPE</i>)	Wavelet entropy (<i>WE</i>)	Wavelet information quantum (<i>WIQ</i>)

mean, variance, standard deviation, root mean square, rectified mean value, waveform factor, pulse factor, crest factor, clearance factor, kurtosis factor, sharpness factor, and entropy.

In the frequency domain, we extracted 13 features based on FFT and WPD. FFT is an algorithm that converts time-domain signals into frequency-domain signals, allowing the decomposition of signals into different frequency components. Vibrational events in optical fibers, such as walking or passing vehicles, exhibit certain periodic characteristics. FFT is suitable for analyzing periodic signals or signals with distinct frequency components. Both the fused OTDR curve of optical fiber pairs and the vibration curve exhibit clear frequency components, making FFT suitable for feature extraction. After FFT, we extracted seven features, as shown in Table II.

In addition to periodic characteristics and distinct frequency components, optical fiber sensing signals also contain irregular and nonstationary frequency components. WPD not only captures more precise frequency information at low frequencies but also captures local temporal characteristics of signals at high frequencies.

In this article, we utilize WPD to divide the original signal into two parts: 1) low frequency and 2) high frequency. Subsequently, each part is further decomposed into low-frequency components (LLP for low-frequency part, LHP for high-frequency part) and high-frequency components (HLP for low-frequency part, HHP for high-frequency part). This results in four frequency components. Table III shows the selected six features after the signal undergoes two-level WPD. Detailed definitions can be found in [45, Table V].

Algorithm 1 PCA

Input: $X \in R^{m \times d}$

Output: $Y \in R^{m \times d}$

- 1: Performs eigen-decomposition on the covariance matrix ($X.X^T$)
- 2: The descending order is used to arrange the Eigen values
- 3: Sort Eigen vectors by Eigen values
- 4: Construct matrix $W(d \times k)$ using the top k ranked Eigen vectors
- 5: Multiplying matrix X by the matrix W yields the reduced-dimensional matrix Y .

C. Feature Dimensionality Reduction

We extracted a total of 28 features in the time and frequency domains. Next, we applied self-difference to the original data and performed feature extraction again. The features extracted from both the original and differential data were concatenated to obtain the required FVs for recognition. The ISU then transmitted the concatenated FVs to the SDN controller.

For different recognition tasks, improving model performance may require different feature sets as FVs. In this work, we compared the performance gains of three different feature dimensionality reduction algorithms in recognition tasks, including principal component analysis (PCA), linear discriminant analysis (LDA), and independent component analysis (ICA).

PCA determines the principal components (PCs) of samples by maximizing the variance of the data. It then projects the data along the direction of maximum variance to transform high-dimensional data into low-dimensional data.

When projecting the original sample set X with d -dimensional features onto a space Y with k -dimensional features, PCA first creates a $d \times k$ transformation matrix W . Next, it performs eigen-decomposition on the covariance matrix ($X.X^T$), followed by sorting the eigenvalues in descending order and selecting the k largest eigenvalues' corresponding eigenvectors [52]. Eigenvalues represent the magnitude of the data, while eigenvectors represent the direction of the data. The steps of PCA are summarized in Algorithm 1, m is the number of data points.

ICA is a linear supervised dimensionality reduction technique. Unlike PCA, which searches for new features along the direction of data variance, ICA seeks directions that maximize the independence of each column in the new data set [53]. The processing steps of ICA are outlined in Algorithm 3. The objective of ICA is to find a transformation matrix W such that each column in the matrix Y obtained by multiplying the data matrix X by W represents independent source signals.

Different from PCA, LDA is a supervised algorithm aimed at maximizing the differences between classes. It extracts k independent features from d features in the source data set. If the data set has c classes, k cannot exceed $c-1$. Algorithm 2 outlines the dimensionality reduction steps of LDA [54]. LDA not only considers the variance of the data but also takes into account the data's classes. First, it computes the within-class scatter matrix S_w and the between-class scatter matrix

Algorithm 2 LDA**Input:** $X \in R^{m \times d}$ **Output:** $Y \in R^{m \times d}$

- 1: Build in-between-class matrix S_b and within-class matrix S_w
- 2: Calculate $S_w^{-1} S_b$
- 3: Sort Eigen vectors by Eigen values in descending order
- 4: To select the eigenvector corresponding to the largest k eigenvalues to form $W(d \times k)$
- 5: Multiplying matrix X by the matrix W yields the reduced-dimensional matrix Y .

Algorithm 3 ICA**Input:** $X \in R^{m \times d}$ **Output:** $Y \in R^{m \times d}$

- 1: Construct matrix $W(d \times k)$
- 2: Let $Y = X \otimes W$
- 3: Measure the independence of each column in Y
- 4: Use optimization algorithms to continually adjust W to get each column in Y is as independent as possible

S_b , which can be obtained, respectively, from

$$S_w = \sum_{i=1}^c \sum_{x \in X_i} (x - \mu_i)(x - \mu_i)^T \quad (5)$$

$$S_b = \sum_{i=1}^c n_i (\mu_i - \mu)(\mu_i - \mu)^T \quad (6)$$

where μ represents the mean vector of all samples, μ_i is the mean vector of the samples in the i th class, n_i is the number of samples in the i th class, and X_i represents the sample set of the i th class. The transformation matrix W consists of the eigenvectors corresponding to the k largest eigenvalues in the matrix $S_w^{-1} S_b$. Transform X using W to obtain the new subspace Y .

IV. RESULTS AND ANALYSIS

Network failures can be sudden, and to further protect business operations from the impact of sudden fiber failures, real-time monitoring and warning of fiber status are necessary. This requires identification models to respond quickly to the network's sudden states. DL, due to its complex model structure and parameters, requires longer response times. In contrast, ML models exhibit good real-time response characteristics. In this section, we validate the performance of different ML models in co-route fiber identification and fiber vibration pattern recognition tasks, including support vector machine (SVM), random forest (RF), naive Bayes (NB), and k -nearest neighbors (KNNs). In this work, 20% of the data was used for validation, while the remaining 80% was used for model training.

A. Co-Cable Recognition

In previous same-cable identification schemes, whether calculating the similarity of OTDR curves for fiber pairs or fusing the FVs of fiber pairs, the feature representation was

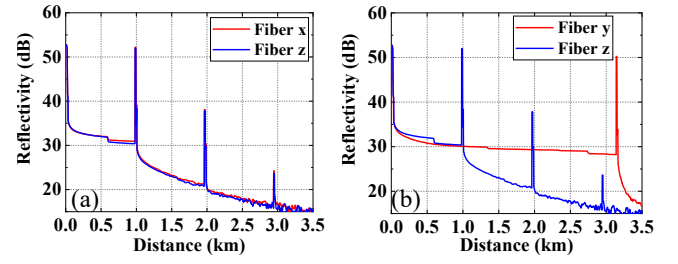


Fig. 5. Schematic of OTDR curves. (a) Co-cable fibers. (b) Non-co-cable fibers.

based on individual fibers. This led to the model ignoring the relationship between fiber pairs, resulting in the inability to further improve the identification accuracy.

Fig. 5 illustrates schematic OTDR curves of co-cable and non-co-cable fibers. In Fig. 5(a), Fiber x and Fiber z are two fibers from the same cable, and their curves are very similar. Fiber y and Fiber z come from different cables, as shown in Fig. 5(b), and the curves of fibers from different cables differ significantly. After differential fusion of the OTDR curves of two fibers to be identified, the fusion curve of the co-cable fiber pair tends to have small values and remains relatively stable overall due to the similarity of the curves of the two fibers, while the fusion curve of the non-co-cable fiber pair exhibits a larger distribution of values.

Motivated by the approach proposed in [56], this article adopts the feature extraction method outlined in [56] as the baseline (FEA-baseline), and then proposes improvements to it, resulting in (FEA-TFW). To the best of our knowledge, this is the first application of time–frequency domain feature extraction in co-route fiber identification. Accuracy, precision, NAR, and F_1 -score are used as performance metrics to evaluate the performance of the proposed FEA, which can be measured using the following formulas [55]:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

$$\text{NAR} = \frac{FN}{TP + FN} \quad (9)$$

$$F_1 - \text{score} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (10)$$

where TP stands for true positives, FN refers to false negatives, FP represents false positives, and TN indicates true negatives.

Table IV presents the simulation results of FEA-baseline and FEA-TFW with different ML algorithms. We applied three feature dimensionality reduction algorithms to the features extracted by FEA-TFW and compared their performance, which is also shown in Table IV.

PCA and ICA selected 16 representative features from the FVs obtained by FEA-TFW, while in the binary classification task, the LDA algorithm could only select one feature. As shown in Table IV, the NB model performed the worst, despite some improvement in accuracy after feature algorithm optimization, the performance of NAR remained poor. The co-cable recognition task focuses more on the recognition

TABLE IV
CO-CABLE FIBER RECOGNITION PERFORMANCE

		Accuracy	Precision	NAR	F1-score
FEA-Baseline	SVM	0.988	0.923	0.199	0.858
	RF	0.929	0.384	0.072	0.543
	KNN	0.991	0.879	0.080	0.899
	NB	0.145	0.051	0	0.096
FEA-TFW	SVM	0.988	0.895	0.160	0.867
	RF	0.958	0.523	0.041	0.677
	KNN	0.995	0.957	0.059	0.949
	NB	0.982	0.894	0.313	0.777
FEA-TFW-PCA	SVM	0.989	0.914	0.156	0.877
	RF	0.996	0.969	0.061	0.954
	KNN	0.996	0.968	0.064	0.951
	NB	0.961	0.549	0.211	0.647
FEA-TFW-LDA	SVM	0.986	0.862	0.177	0.842
	RF	0.978	0.750	0.214	0.768
	KNN	0.986	0.87	0.196	0.836
	NB	0.982	0.761	0.128	0.813
FEA-TFW-ICA	SVM	0.995	0.988	0.105	0.939
	RF	0.997	0.984	0.053	0.965
	KNN	0.995	0.973	0.081	0.945
	NB	0.969	0.638	0.238	0.695

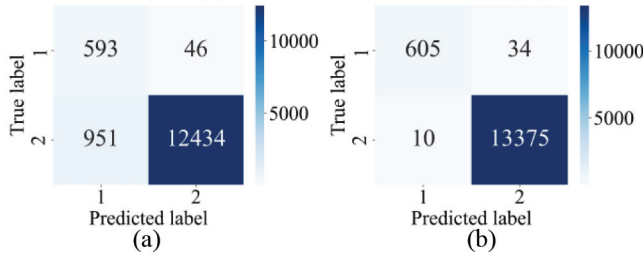


Fig. 6. Confusion matrices of the RF model in (a) FEA-Baseline and (b) FEA-TFW-ICA for co-cable recognition.

performance of co-cable fibers, but the proportion of co-cable fiber pairs in the total fiber is very small. The NB model tends to lean toward the higher proportion of non-co-cable fibers, resulting in inaccurate judgments of co-cable fibers.

After dimensionality reduction using the ICA algorithm, the RF model showed the best performance, achieving a recognition accuracy of 0.997, an F1-score of 0.965, and significantly reducing the NAR to 0.053. Although the NAR performance was optimal with a score of only 0.041 in the RF model with FEA-TFW, the other three indicators were not as good as the performance of FEA-TFW-ICA. The NB model had poor overall performance in FEA-Baseline, and although the NAR was 0, it is not further discussed.

In Fig. 6, the confusion matrix for the RF model in FEA-Baseline and FEA-TFW-ICA is presented. From Fig. 6, we can observe that there are only 639 pairs of co-cable fibers used for testing, while the number of non-co-cable fiber pairs reaches 13 380. When FEA-Baseline was used, 46 pairs of the same fiber were not correctly identified, with an NAR of 7.2%. With FEA-TFW-ICA, the NAR dropped to 4.56%. The data used in our experiments were collected from the communication network of the operator, involving typical

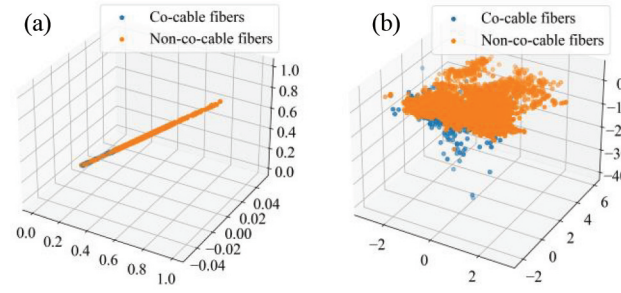


Fig. 7. Feature visualization of (a) FEA-Baseline and (b) FEA-TFW-ICA.

network scenarios, such as metropolitan networks, backbone networks, and aggregation networks. The network comprises over hundreds of thousands of optical cables, with a limited number of fibers per cable. The proportion of coiled fibers at 4.56% aligns with the actual situation in the network. Among the 639 pairs of co-cable fibers tested, the RF model based on FEA-TFW-ICA successfully identified 605 pairs with an outstanding performance, yielding an NAR of only 0.053.

To better visualize the performance advantage of FEA-TFW-ICA, we present the visual distribution of FVs extracted from FEA-Baseline and FEA-TFW-ICA in Fig. 7(a) and (b), respectively. It can be observed that FEA-TFW-ICA effectively separates more co-cable fibers from non-co-cable fibers, which contributes to the classification performance of the model.

In our previous work, we proposed a two-layered cascaded RF algorithm for co-cable recognition, achieving an accuracy of 0.95 and an F1-score of 0.9. Based on FEA-TFW-ICA, the RF model's performance in co-cable recognition is excellent, surpassing our previous work.

B. Trench Characterization

In the infrastructure of networks, multiple optical fibers are often laid in the same trenches. When facing geological disasters or human excavation, there is a risk of fiber breakage for all fibers in the same trench. To identify the fibers in the same trench, namely, co-trench fibers, we collected a large amount of fiber vibration data in real network scenarios from the telecommunications operator's network. The process of obtaining vibration data is illustrated in Fig. 4.

Fig. 8 shows the vibration curves extracted from two fibers. In our experiment, the time length of the vibration curve is 2000 ms, with a spatial resolution of 3.2 m.

The experiment simulates a real online acquisition environment, where four fibers in three ducts are simultaneously tested (two fibers in one duct are measured). A total of 3019 vibration curves were extracted during the test period. In the simulation, the ratio of training data set to validation data set is 8:2.

Table V displays the accuracy performance of various ML models on different feature extraction methods. Similar to the co-cable identification simulation, PCA and ICA selected 16 representative features from the FEA-TFW FVs. However, due to the presence of three ducts in the validation data set, the number of features extracted by the LDA algorithm was set

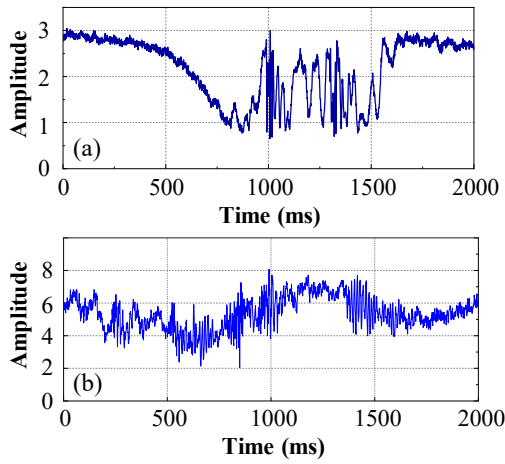


Fig. 8. Schematic of the fiber vibration curves.

TABLE V
COMPARISON OF THE ACCURACY OF TRENCH CLASSIFICATION

	SVM	RF	KNN	NB
FEA-Baseline	0.958	0.952	0.958	0.935
FEA-TFW	0.962	0.957	0.958	0.94
FEA-TFW-PCA	0.967	0.977	0.968	0.95
FEA-TFW-LDA	0.967	0.96	0.965	0.95
FEA-TFW-ICA	0.965	0.963	0.962	0.75

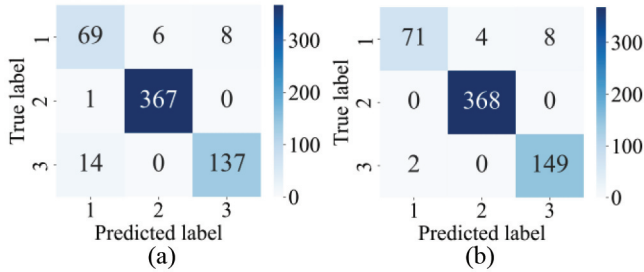


Fig. 9. Confusion matrices of the RF model in (a) FEA-Baseline and (b) FEA-TFW-PCA for trench classification.

to 2. It can be observed that the FVs after PCA dimensionality reduction achieved the best accuracy performance among all ML models. Specifically, the RF model based on FEA-TFW-PCA achieved a recognition accuracy of 0.977, which is the best performance observed in all tests.

Fig. 9 illustrates the confusion matrix of the RF model based on FEA-Baseline and FEA-TFW-PCA. It can be observed that after FEA-TFW-PCA is used, all samples in trench2 are correctly classified, and the number of samples misclassified as trench1 in trench3 is reduced from 14 to 2. In contrast, trench1 exhibits a higher NAR, which is attributed to the lower number of valid vibration curves collected from trench1. The performance metrics of various ML models based on FEA-TFW-PCA in each duct are presented in Table VI. Despite the NAR of trench1 being 0.145, its F1-score is 0.91, which is the best performance among all models.

The effectiveness of FEA-TFW-PCA in trench classification is visually demonstrated in Fig. 10. Fig. 10(a) shows the visualization of FVs extracted by FEA-Baseline. After feature

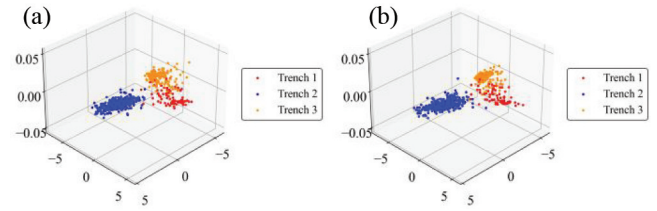


Fig. 10. Trench feature visualization of (a) FEA-Baseline and (b) FEA-TFW-PCA.

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT MODELS IN TRENCH CLASSIFICATION WITH FEA-TFW-PCA

	Trench	Precision	NAR	F1-score
SVM	1	0.899	0.145	0.877
Accuracy=0.967	2	0.992	0.000	0.996
	3	0.941	0.053	0.944
RF	1	0.973	0.145	0.910
Accuracy=0.977	2	0.989	0.000	0.995
	3	0.949	0.013	0.968
KNN	1	0.901	0.120	0.890
Accuracy=0.968	2	0.995	0.008	0.993
	3	0.942	0.040	0.951
NB	1	0.807	0.145	0.830
Accuracy=0.950	2	0.989	0.016	0.986
	3	0.939	0.079	0.93

extraction with FEA-TFW-PCA, the three ducts are distinctly separated in the feature space, as depicted in Fig. 10(b).

The previous task of identifying co-trench fibers aimed to distinguish whether any two fibers belong to the same trench. In contrast, the architecture proposed in this article can accurately identify which trench a fiber belongs to, thereby distinguishing whether any two fibers belong to the same trench. This represents a significant advancement in co-trench fiber identification, which will save time and costs for fiber maintenance.

C. Fiber Vibration Pattern Recognition

Fiber fracture can significantly impact the survivability of business operations. Identifying co-route fibers can mitigate the risk of service interruption in already laid fibers and prevent co-route fiber phenomenon occurrence in future fiber deployments. However, co-route fiber identification can only identify potential risks before and after fiber deployment, and cannot provide early warnings for sudden fiber fractures. In this article, we use an ISAC architecture to monitor the real-time status of fibers. When vibration events threatening the fiber status, such as digging or knocking, are detected, timely alerts are sent to the SDN controller. This enables network administrators to promptly adjust services on the alerted fibers, thereby avoiding service interruptions.

The fiber vibration data set utilized in this study was sourced from the open data set of ϕ -OTDR events, as provided in [56]. This data set comprises a total of 15 612 samples, encompassing six distinct event types: background noise, digging, knocking, watering, shaking, and walking, denoted as 1, 2, 3,

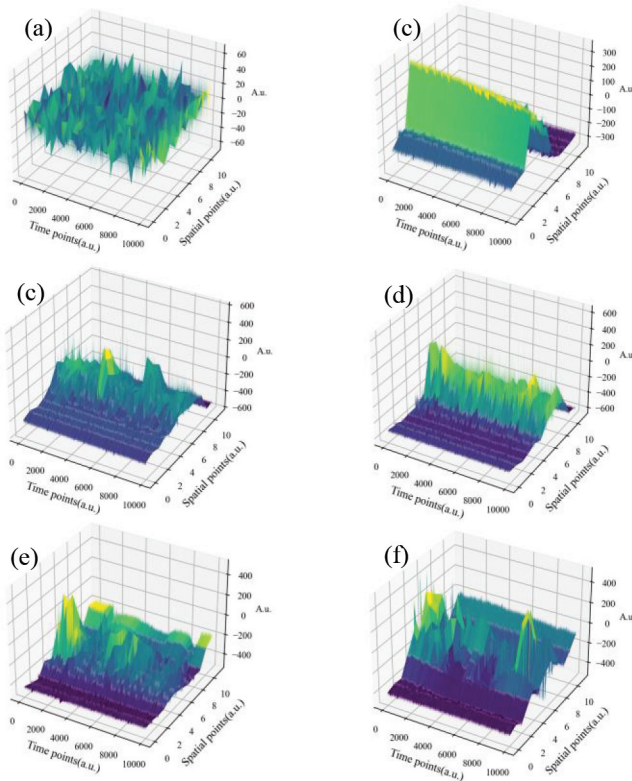


Fig. 11. Spatial-temporal samples of the six typical events. (a) Background noise. (b) Digging. (c) Knocking. (d) Watering. (e) Shaking. (f) Walking.

4, 5, and 6, respectively. Each individual sample is structured as a matrix of 10000 rows and 12 columns, representing 10000 time points and 12 spatial locations, respectively. Fig. 11 illustrates 3-D waterfall plots of samples representing the aforementioned six event types. Herein, the horizontal and vertical axes denote temporal and spatial points, while the depth axis depicts the dimensionless vibration intensity at corresponding spatiotemporal positions.

Reference [56] also provides two baseline models, namely, SVM and 2D-CNN (convolutional neural network, 2-D approach). In this study, the FEA used by the SVM model is referred to as FEA-baseline, while the improved FEA is denoted as FEA-TFW. FEA-baseline divides the original samples into 12 channels based on spatial locations, and extracts 32 features from both the original and differential signals for each channel. Consequently, the resulting FV has a length of 384 (32*12). However, longer FVs can lead to larger search spaces and affect model performance.

Therefore, feature dimensionality reduction algorithms were applied to both FEA-baseline and FEA-TFW, and their accuracy performances on different ML models are presented in Table VII.

The two baseline models mentioned in [36]—SVM and 2D-CNN—achieved recognition accuracies of 0.826 and 0.940, respectively. Following feature extraction with FEA-TFW-LDA, the accuracy of various ML models reached their optimum, with the SVM model performing the best, achieving an accuracy of 0.980. This performance surpassed that of the two baseline models. Precision, NAR, and f1 scores for SVM

TABLE VII
PERFORMANCE OF FIBER OPTIC VIBRATION EVENT CLASSIFICATION

	SVM	RF	KNN	NB
FEA-Baseline	0.826	0.936	0.858	0.486
FEA-Baseline-PCA	0.905	0.931	0.860	0.790
FEA-Baseline-LDA	0.91	0.924	0.925	0.929
FEA-Baseline-ICA	0.959	0.918	0.818	0.723
FEA-TFW	0.812	0.962	0.895	0.240
FEA-TFW-PCA	0.930	0.927	0.899	0.819
FEA-TFW-LDA	0.980	0.976	0.978	0.974
FEA-TFW-ICA	0.906	0.885	0.662	0.747

TABLE VIII
PERFORMANCE OF DIFFERENT METRICS IN EACH EVENT CATEGORY

	Events	Precision	NAR	F ₁ -score
SVM	1	0.987	0.014	0.986
	2	0.974	0.028	0.973
FEA-TFW-LDA	3	0.971	0.031	0.970
	4	0.991	0.011	0.990
Accuracy=0.980	5	0.991	0.011	0.990
	6	0.965	0.037	0.964

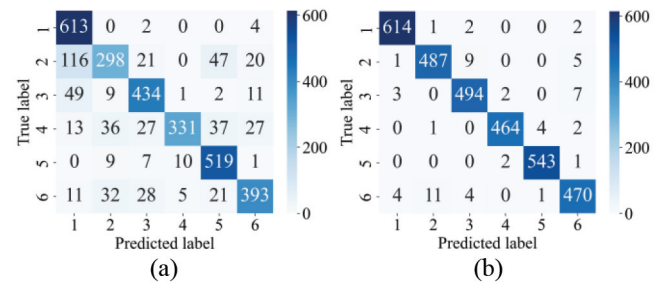


Fig. 12. Confusion matrices of the SVM model for fiber vibration event classification in (a) FEA-Baseline and (b) FEA-TFW-LDA.

with FEA-TFW-LDA can be found in Table VIII, where it is evident that the false alarm rates for each event are very low.

Fig. 12 illustrates the confusion matrix for SVM with FEA-Baseline and FEA-TFW-LDA. It can be observed that before using FEA-TFW-LDA, there are more cases of misclassification of the model, but after using FEA-TFW-LDA, the number of misclassification of each category of data is significantly reduced. In Fig. 12(b), the majority of samples are correctly classified, while there are minor instances of confusion between the events digging, knocking, and walking. This phenomenon arises due to the sudden application of stress on the fiber for these three types of events (walking events include a single stampede event), making them prone to confusion when the external forces are similar.

To visually demonstrate the superiority of FEA-TFW-LDA over FEA-baseline, we conducted a visualization of the FVs extracted by both methods in Fig. 13. We utilized the LDA algorithm to reduce the dimensionality of the extracted FVs to three dimensions for ease of mapping in the 3-D space. It can be observed that in Fig. 13(a), the distribution of FVs extracted by FEA-baseline is relatively concentrated at various event points. Conversely, FEA-TFW-LDA exhibits

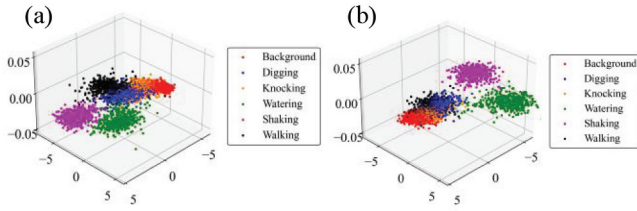


Fig. 13. Fiber vibration event feature visualization of (a) FEA-Baseline and (b) FEA-TFW-LDA.

better visualization effects, enabling better differentiation of different event points in the feature space.

V. CONCLUSION

In this article, we propose the ISAC architecture for transport networks, leveraging live fiber data collected from telecommunications operators to train ML models. This architecture achieves real-time identification of co-route fibers and fiber state alerts. Notably, in co-cable fiber identification, we introduce for the first time the fusion of OTDR curves from fiber pairs as the fundamental unit for feature extraction, achieving a remarkable NAR of 5.3% and a recognition accuracy of 99.7%. In the task of co-trench fiber identification, we achieve a high accuracy of 97.7% through the novel delineation of fiber trenches. Furthermore, in fiber state prediction, our solution enhances the recognition accuracy from the baseline model's 82.6%–98% using publicly open fiber vibration data sets. Extensive simulation experiments have proved the superior performance of the proposed architecture, which is a great progress in transport network operation and maintenance, and provides a reliable scheme for further guaranteeing the survivability of everything interconnection in 6G IoE.

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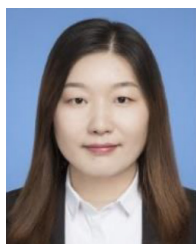
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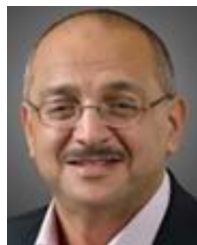
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